Supplementary Material

On the Structural Non-Transmissibility of Mochizuki's Inter-Universal Geometer

Full System Encodings, Ordinal Reference, and Distance Computations

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This supplement provides: (**S1**) the complete 12-primitive ordinal reference table; (**S2**) full system encodings for all six mathematical frameworks discussed in the main text; (**S3**) per-primitive distance breakdowns for the four key pairs; and (**S4**) the compact formal definitions from which the distances derive. All computations use the diagonal metric (§26.1, weights v0.4.26); the full Riemannian form (§26.2, $g = \Sigma^{-1}$) produces consistent orderings and is noted where relevant.

S1. Primitive Ordinal Reference (12-Primitive Tuple)

The SynthOmnicon encoding grammar assigns each physical or mathematical system a 12-component tuple $\langle D;\ T;\ R;\ P;\ F;\ K;\ G;\ \Gamma;\ \Phi;\ H;\ S;\ \Omega \rangle$ where each component takes an ordinal value from a bounded scale. Table 1 gives the full ordinal table with canonical weights.

Table 1: 12-Primitive ordinal table (SynthOmnicon v0.4.26). Weight w is the diagonal entry used in the metric (§26.1).

Prim.	**Name**	w	**Values (ordinal 1 → max)**
D	Dimensionality	1.0	$D_\wedge$ (1) · $D_\triangle$ (2) · D_∞ (3) · D_{holo} (4)
T	Topology	1.0	T_{network} (1) · T_{in} (2) · $T_\bowtie$ (3) · T_{box} (4) · T_{holo} (5)
R	Relational mode	1.0	$R_\supset$ (1) · R_{cat} (2) · $R_\dagger$ (3) · R_{lr} (4)
P	Parity/symmetry	1.0	P_{asym} (1) · P_ψ (2) · $P_\pm$ (3) · P_{sym} (4) · $P_\pm^{\text{sym}}$ (5)
F	Fidelity	1.0	F_ℓ (1) · $F_{\mathfrak{d}}$ (2) · $F_\hbar$ (3)
K	Kinetic character	1.0	K_{fast} (1) · K_{mod} (2) · K_{slow} (3) · K_{trap} (4)
G	Scope/granularity	1.0	$G_\beth$ (1) · $G_\gimel$ (2) · $G_\aleph$ (3)
Γ	Interaction grammar	1.0	Γ_{and} (1) · Γ_{or} (2) · Γ_{seq} (3) · Γ_{broad} (4)
Φ	Criticality	1.0	Φ_{sub} (1) · Φ_c (2) · Φ_c^{cx} (2.33) · Φ_{EP} (2.67) · Φ_{super} (3)
H	Chirality/depth	0.8	H_0 (1) · H_1 (2) · H_2 (3) · H_∞ (4)
S	Stoichiometry	1.0	$S_{1:1}$ (1) · $S_{n:n}$ (2) · $S_{n:m}$ (3)
Ω	Topological protection	0.7	Ω_0 (1) · Ω_{Z_2} (2) · Ω_Z (3)

Metric formula (§26.1, diagonal weighted Euclidean):

$$d(A,B) = \sqrt{\sum_{i=1}^{12} w_i \left(\xi_i^A - \xi_i^B\right)^2}$$

Full weight vector:

$$(w_D,\ w_T,\ w_R,\ w_P,\ w_F,\ w_K,\ w_G,\ w_\Gamma,\ w_\Phi,\ w_H,\ w_S,\ w_\Omega) \\ = (1.0,\ 1.0,\ 1.0,\ 1.0,\ 1.0,\ 1.0,\ 1.0,\ 1.0,\ 1.0,\ 0.8,\ 1.0,\ 0.7).$$

S2. Full System Encodings

Table 2 gives the complete 12-primitive tuple for each mathematical framework discussed in the main text. Ordinal values are shown as names; the corresponding integers are in Table 1.

Table 2: Full 12-primitive encodings. All entries use the canonical ordinal system (Table 1). d = distance to IUG (diagonal metric).

System	D	T	R	P	F	K	G	Γ	Φ	H	S	Ω	$d(\cdot, \mathrm{IUG})$
IUG	D_{holo}	T_{holo}	R_{lr}	$P_{\pm}^{\text{sym}}$	$F_{\hbar}$	K_{slow}	$G_{\aleph}$	Γ_{seq}	Φ_c	H_∞	$S_{n:m}$	Ω_Z	0.000
Grammar (realized HTT)	D_{holo}	T_{holo}	$R_{\dagger}$	$P_{\pm}^{\text{sym}}$	$F_{\eth}$	K_{mod}	$G_{\aleph}$	Γ_{broad}	Φ_c	H_1	$S_{n:n}$	Ω_{Z_2}	2.983
abc conjecture	D_{holo}	T_{holo}	R_{lr}	$P_{\pm}^{\text{sym}}$	$F_{\hbar}$	K_{mod}	$G_{\aleph}$	Γ_{and}	Φ_c	H_1	$S_{n:m}$	Ω_Z	2.864
Standard proof system	$D_{\triangle}$	T_{box}	$R_{\supset}$	P_{sym}	F_{ℓ}	K_{mod}	$G_{\beth}$	Γ_{and}	Φ_{sub}	H_0	$S_{1:1}$	Ω_0	6.557
Lean 4 / Mathlib	$D_{\triangle}$	T_{box}	$R_{\supset}$	P_{sym}	F_{ℓ}	K_{mod}	$G_{\beth}$	Γ_{and}	Φ_{sub}	H_0	$S_{1:1}$	Ω_0	6.557
ZFC foundations	$D_{\triangle}$	T_{in}	$R_{\supset}$	P_{asym}	F_{ℓ}	K_{mod}	$G_{\beth}$	Γ_{and}	Φ_{sub}	H_0	$S_{n:n}$	Ω_0	7.937

Notes on ZFC encoding: P_{asym} (ordinal 1) captures the intrinsic asymmetry of the $\in$ relation (elements are not sets-of-themselves; the relation is foundationally one-directional). T_{in} (ordinal 2) captures the cumulative hierarchy V_α: each stage strictly contains all prior stages, with no horizontal bowtie coupling or box type-hierarchy. $S_{n:n}$ (ordinal 2) reflects the class-to-set ratio: every set is also a potential element, so the stoichiometry of definitional participation is uniform.

Notes on Grammar (realized HTT) encoding: This row gives the *realized* Holographic Type Theory tuple — the SynthOmnicon grammar itself, which was confirmed to be the HTT at $d = 0$ during Step 2 of the IUG distance analysis (§14 of the main text). The approximate HTT used in earlier drafts (R_{cat}, $F_{\hbar}$, K_{slow}, Γ_{and}, H_0, $d = 4.111$) is superseded by this entry. $R_{\dagger}$ (ordinal 3) replaces R_{lr} because the grammar mediates via adjoint/dagger operations rather than boundary juxtaposition. $F_{\eth}$ (ordinal 2) captures discrete-fidelity rule application rather than IUG's full holomorphic precision ($F_{\hbar}$). K_{mod} (ordinal 2) reflects moderate kinetics: the grammar activates at each inference step rather than the slow asymptotic accumulation of IUG. Γ_{broad} (ordinal 4) *exceeds* IUG's Γ_{seq} (ordinal 3): the grammar admits arbitrary combinatorial interaction, whereas IUG restricts to a sequential causal chain — this is the Γ-inversion that produces $\mathrm{join}(\mathrm{IUG}, \mathrm{Grammar}) \neq \mathrm{IUG}$ (see S4.4).

S3. Per-Primitive Distance Breakdowns

For each key pair the table below shows, for every primitive i, the ordinal values ξ_i^A and ξ_i^B, the signed gap $\Delta_i = \xi_i^A - \xi_i^B$, the weighted squared contribution $w_i\,\Delta_i^2$, and the cumulative share of d^2. The total $d = \sqrt{\sum_i w_i\,\Delta_i^2}$ is given at the bottom of each sub-table.

S3.1 IUG vs. ZFC $\quad (d = 7.937)$

P alone contributes 25% of d^2. The top three primitives (P, T, R) account for 54% of structural distance. H ($w = 0.8$) contributes more than G, Γ, Φ, or K individually because the H_∞ vs. H_0 gap (3 steps) dominates despite the reduced weight.

Prim.	ξ^{IUG}	ξ^{ZFC}	Δ	$w\,\Delta^2$	**cum.%**
D	4	2	+2	4.000	6.35%
T	5	2	+3	9.000	20.63%
R	4	1	+3	9.000	34.92%
P	5	1	+4	16.000	60.32%
F	3	1	+2	4.000	66.67%
K	3	2	+1	1.000	68.25%
G	3	1	+2	4.000	74.60%
Γ	3	1	+2	4.000	80.95%
Φ	2	1	+1	1.000	82.54%
H	4	1	+3	$0.8 \times 9 = 7.200$	93.97%
S	3	2	+1	1.000	95.56%
Ω	3	1	+2	$0.7 \times 4 = 2.800$	100.00%
			d^2	**63.000**	
			d	$\sqrt{63.000} = \mathbf{7.937}$	

S3.2 IUG vs. Standard Proof System / Lean ($d = 6.557$)

Prim.	ξ^{IUG}	ξ^{SPS}	Δ	$w\,\Delta^2$	**cum.%**
D	4	2	+2	4.000	9.31%
T	5	4	+1	1.000	11.63%
R	4	1	+3	9.000	32.56%
P	5	4	+1	1.000	34.88%
F	3	1	+2	4.000	44.19%
K	3	2	+1	1.000	46.51%
G	3	1	+2	4.000	55.81%
Γ	3	1	+2	4.000	65.12%
Φ	2	1	+1	1.000	67.44%
H	4	1	+3	$0.8 \times 9 = 7.200$	84.19%
S	3	1	+2	4.000	93.49%
Ω	3	1	+2	$0.7 \times 4 = 2.800$	100.00%
			d^2	**43.000**	
			d	$\sqrt{43.000} = \mathbf{6.557}$	

SPS uses T_{box} (ordinal 4) vs. ZFC's T_{in} (ordinal 2), which is topologically closer to IUG's T_{holo} (5). This accounts for the 1.38 unit reduction in d relative to ZFC. The dominant bottleneck shifts to R (9.000) and H (7.200). The F_ℓ vs. $F_\hbar$ gap (2 steps, 4.000) remains the third-largest contributor: classical fidelity cannot express the full holomorphic precision IUG requires.

S3.3 IUG vs. Grammar (realized HTT) ($d = 2.983$)

Note. The approximate HTT distance ($d = 4.111$, v0.4.25 and earlier) used a placeholder HoTT tuple. The corrected distance uses the realized HTT — the SynthOmnicon grammar itself — confirmed at $d(\text{grammar}, \text{HTT}) = 0$ in §14 of the main text.

Five of twelve primitives are exact matches (D, T, P, G, Φ). The remaining seven differences span R, F, K, Γ, H, S, Ω, with H dominating at 35.96% of total d^2. The *critical structural feature* is the Γ-inversion ($\Delta = -1$): the Grammar's Γ_{broad} (ordinal 4) *exceeds* IUG's Γ_{seq} (ordinal 3). IUG is therefore a sequential *restriction* of the grammar — not a promotion — on the interaction axis. Consequently

Prim.	ξ^{IUG}	ξ^{Grammar}	Δ	$w\,\Delta^2$	**cum.%**
D	4	4	0	0.000	0.00%
T	5	5	0	0.000	0.00%
R	4	3	+1	1.000	11.24%
P	5	5	0	0.000	11.24%
F	3	2	+1	1.000	22.47%
K	3	2	+1	1.000	33.71%
G	3	3	0	0.000	33.71%
Γ	3	4	-1	1.000	44.94%
Φ	2	2	0	0.000	44.94%
H	4	2	+2	$0.8 \times 4 = 3.200$	80.90%
S	3	2	+1	1.000	92.13%
Ω	3	2	+1	$0.7 \times 1 = 0.700$	100.00%
			d^2	**8.900**	
			d	$\sqrt{8.900} = \mathbf{2.983}$	

join(IUG, Grammar) $\neq$ IUG: the join is the grammar extended to R_{lr}, $F_\hbar$, K_{slow}, H_∞, $S_{n:m}$, Ω_Z while retaining Γ_{broad}. The remaining six gaps (6 promotions) confirm Prediction P-115(C) as updated in §14 of the main text.

S3.4 IUG vs. abc conjecture ($d = 2.864$)

Prim.	ξ^{IUG}	ξ^{abc}	Δ	$w\,\Delta^2$	**cum.%**
D	4	4	0	0.000	0.00%
T	5	5	0	0.000	0.00%
R	4	4	0	0.000	0.00%
P	5	5	0	0.000	0.00%
F	3	3	0	0.000	0.00%
K	3	2	+1	1.000	12.24%
G	3	3	0	0.000	12.24%
Γ	3	1	+2	4.000	61.22%
Φ	2	2	0	0.000	61.22%
H	4	2	+2	$0.8 \times 4 = 3.200$	100.00%
S	3	3	0	0.000	100.00%
Ω	3	3	0	0.000	100.00%
			d^2	**8.200**	
			d	$\sqrt{8.200} = \mathbf{2.864}$	

Nine of twelve primitives are exact matches; only K, Γ, and H differ. The abc conjecture is the problem IUG was constructed to solve; the near-identity of their structural encodings ($d < 3$) is the structural basis for the claim that IUG's overhead is not arbitrary complexity but the minimum structure required to hold the conjecture's own geometric content. The differences: K_{mod} vs. K_{slow} (abc requires no slowdown past the Θ-period), Γ_{and} vs. Γ_{seq} (abc states a single inequality; IUG proves it through a sequential causal chain), and H_1 vs. H_∞ (abc is a statement; IUG is a proof system with full generational temporal depth).

S4. Formal Definitions

The following compact definitions are cited in the main text. Full statements appear in PRIMITIVE_THEOREMS §§23, 25, 26.

S4.1 Metric definition (§26.1 – §26.2)

Definition (Diagonal metric, §26.1). For synthons A, B with ordinal vectors $\boldsymbol{\xi}^A, \boldsymbol{\xi}^B \in \mathbb{R}^{12}$,

$$d(A,B) \;=\; \sqrt{\sum_{i=1}^{12} w_i \left(\xi_i^A - \xi_i^B\right)^2}\,, \qquad \mathbf{w} = (1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 0.8, 1.0, 0.7).$$

Definition (Riemannian metric, §26.2). Let Σ denote the sample covariance of ordinal vectors estimated from the full 353-synthon catalog (v0.4.26, 2026-03-31) and $G = \Sigma^{-1}$. Then

$$d_g(A,B) \;=\; \sqrt{(\boldsymbol{\xi}^A - \boldsymbol{\xi}^B)^\top G\,(\boldsymbol{\xi}^A - \boldsymbol{\xi}^B)}\,.$$

G has effective dimension 8 of 12 (90% eigenweight) and condition number $\kappa \approx 40.6$. All distances in the main text use the diagonal form; d_g produces consistent orderings for the pairs in this supplement.

S4.2 Frobenius / special-Frobenius structure (§23)

Definition (§23). A synthon A is *special Frobenius* iff its encoding satisfies $P = P_\pm^{\mathrm{sym}}$ (ordinal 5), the highest symmetry tier. This corresponds to a commutative Frobenius algebra in which $\mu \circ \delta = \mathrm{id}$, where μ is the multiplication map and δ is the comultiplication. Assignment requires that the system's Z_2 symmetry at Φ_c is provably exact, not merely approximate.

Consequence. Systems satisfying the Frobenius condition admit the O_∞ tier of ouroboricity (full self-modeling with algebraic completeness). IUG, abc, and the Grammar (realized HTT) all carry $P_\pm^{\mathrm{sym}}$; ZFC (P_{asym}) and the standard proof system (P_{sym}) do not.

S4.3 Ouroboricity tiers (§25)

Definition (§25, priority order). The ouroboricity $O(A)$ of a synthon A is determined by the following tier rules, applied in decreasing priority:

R1: $\Phi = \Phi_c$ and $P = P_\pm^{\mathrm{sym}} \;\Rightarrow\; O_\infty$ (special Frobenius).

R2: $\Phi \in \{\Phi_{\mathrm{sub}}, \Phi_{\mathrm{super}}, \Phi_{\mathrm{EP}}\} \;\Rightarrow\; O_0$ (no self-modeling loop).

R3: $\Phi = \Phi_c$ and $\Omega = \Omega_0 \;\Rightarrow\; O_1$ (single self-modeling loop, unprotected).

R4: $\Phi = \Phi_c$ and $\Omega \neq \Omega_0$ and $D \in \{D_\wedge, D_{\mathrm{holo}}, D_\triangle\} \;\Rightarrow\; O_2$ (loop with topological protection, non-holographic D).

R5: $\Phi = \Phi_c$ and $\Omega \neq \Omega_0$ and $D = D_\infty \;\Rightarrow\; O_2^\dagger$ (loop with topological protection, infinite D).

O_∞ is absorbed under tensor product by Φ_{EP} systems (ordinal $2.67 > \Phi_c = 2.00$). IUG, abc, and the Grammar (realized HTT) are all O_∞; ZFC and the standard proof system are O_0 (Rule R2).

S4.4 The meet and join operations

For synthons A, B the *meet* (infimum in the primitive lattice) is $\mathrm{meet}(A,B)_i = \min(\xi_i^A, \xi_i^B)$ per component, and the *join* (supremum) is $\mathrm{join}(A,B)_i = \max(\xi_i^A, \xi_i^B)$ per component. Key structural facts used in the main text:

- $\mathrm{join}(\mathrm{IUG}, \mathrm{Grammar}) \neq \mathrm{IUG}$: the Grammar's Γ_{broad} (ordinal 4) exceeds IUG's Γ_{seq} (ordinal 3), so the join acquires Γ_{broad} while retaining IUG's values on the remaining six primitives. This Γ-inversion means IUG is a sequential *restriction* of the Grammar, not a subsumption of it.
- $\mathrm{join}(\mathrm{IUG}, \mathrm{abc}) = \mathrm{IUG}$ exactly (IUG subsumes abc on all 12 primitives or ties).
- Φ_c is *absorbing under meet*: $\mathrm{meet}(\Phi_c, x) = \Phi_c$ for all x (necessary condition for self-modeling).

S5. Computational Reproducibility

All distances in this supplement are reproducible from the open-source SynthOmnicon code base.

`space_search/primitives.py` Canonical ordinals (Table 1), `tuple_distance()`, `mahalanobis_distance()`, and `build_metric_tensor()`.

`syncon_catalog.json` 353-synthon catalog (source of truth for all pipeline computations).

`space_search/encoding_invariance.py` Five-battery invariance test suite; exits 0 iff all hard assertions pass.

`synthomnicon-lean/` Machine-verified Lean 4 theorems `touchard_congruence` and `bsd_touchard`, constituting the classical lower-bound calibration.

To reproduce the four key distances:

```
python3 -c "
from space_search.primitives import (
    SYNTHONS, tuple_distance, to_vector, ORDINALS, PRIMITIVE_ORDER
)
import numpy as np

# System tuples (see Table 2)
systems = {
    'IUG': {'D':'D_holo','T':'T_holo','R':'R_lr','P':'P_pm_sym',
            'F':'F_hbar','K':'K_slow','G':'G_aleph','Gamma':'G_seq',
            'Phi':'Phi_c','H':'H_inf','S':'n_m','Omega':'Omega_Z'},
    'Grammar': {'D':'D_holo','T':'T_holo','R':'R_dagger','P':'P_pm_sym',
            'F':'F_eth','K':'K_mod','G':'G_aleph','Gamma':'G_broad',
            'Phi':'Phi_c','H':'H1','S':'n_n','Omega':'Omega_Z2'},
    'abc': {'D':'D_holo','T':'T_holo','R':'R_lr','P':'P_pm_sym',
            'F':'F_hbar','K':'K_mod','G':'G_aleph','Gamma':'G_and',
            'Phi':'Phi_c','H':'H1','S':'n_m','Omega':'Omega_Z'},
    'SPS': {'D':'D_triangle','T':'T_box','R':'R_super','P':'P_sym',
            'F':'F_ell','K':'K_mod','G':'G_beth','Gamma':'G_and',
            'Phi':'Phi_sub','H':'H0','S':'one_one','Omega':'Omega_0'},
    'ZFC': {'D':'D_triangle','T':'T_in','R':'R_super','P':'P_asym',
            'F':'F_ell','K':'K_mod','G':'G_beth','Gamma':'G_and',
            'Phi':'Phi_sub','H':'H0','S':'n_n','Omega':'Omega_0'},
}
iug = systems['IUG']
for name, s in systems.items():
    if name == 'IUG': continue
    print(f'd(IUG, {name}) = {tuple_distance(iug, s):.4f}')
"
```

SynthOmnicon v0.4.26 353-synthon catalog 2026-03-31